

Advanced Development Tools for Deep Learning and AI

Aldebaro Klautau
Federal University of Pará (UFPA) / LASSE

Computational Intelligence Class

Tensorboard, Optuna, Model distillation, Quantization tools,
Huggingface

November 07, 2026

Machine and deep learning frameworks



DL training frameworks implement sophisticated algorithms, which use “backpropagation”, automatic differentiation and stochastic gradient descent (SGD)



Most used language



Facebook's / Meta's



&



Google's, TF versions 1 and 2, with high level Keras API

Deployment frameworks: facilitate pruning the models and quantizing the weights for acceleration

Qualcomm's AI Model Efficiency Toolkit

www.tinymml.org



TorchScript, Tensorflow Lite & PyTorch Quantization



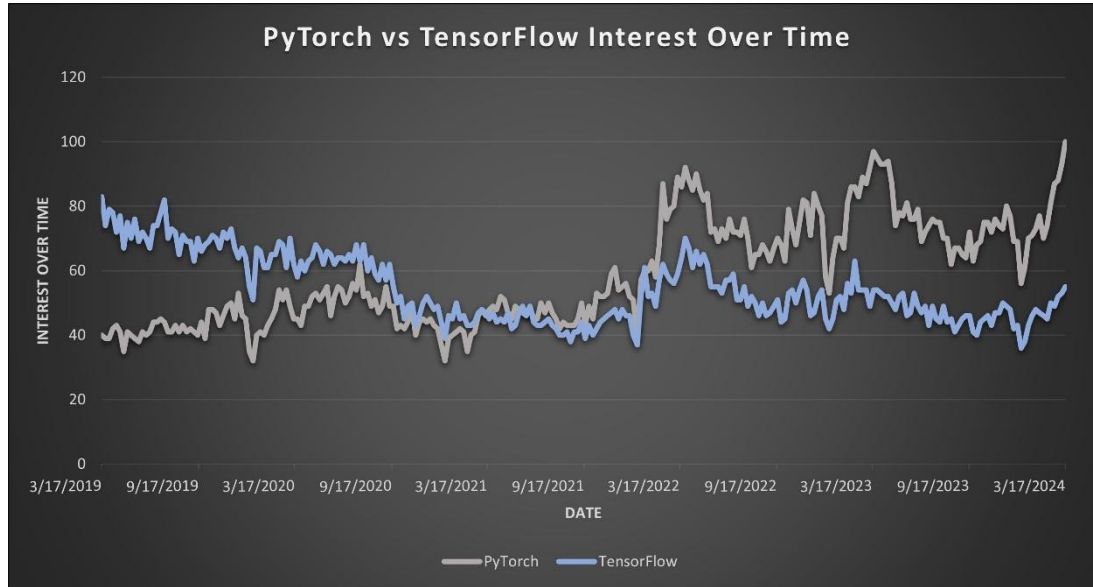
Other tools: NVIDIA, Intel, etc.

Auxiliary tools for (shallow) machine learning, debugging, assessing models and running on cloud



It may not be trivial to set up your development workflow

*Tensorflow 2 versus Pytorch



TF2: Easier deployment to cloud, servers, mobile, and IoT devices: TensorFlow Serving and TensorFlow Lite

Pytorch: More models, for instance, at <https://huggingface.co/>

- 85% only PyTorch
- 8% only TF
- 7% both

Tools such as:

<https://github.com/Lyken17/pytorch-OpCounter> to count MACs / FLOPs